

THE HONG KONG UNIVERSITY OF SCIENCE & TECHNOLOGY
ARIN5204

Yang, Fengshuo/21216054

March 17, 2026

Quiz 4

1 Deadly Triad :

This is a phenomenon of instability in reinforcement learning caused by the simultaneous occurrence of three specific conditions.

1.1 three conditions

- Bootstrapping: means self-reinforcement. Using its own estimate to update value, rather than the actual return at the end of the episode..
- Off-policy Learning: means updating the target policy with other policies.
- Function Approximation: leads to approximation error.

1.2 result

- Errors accumulate and explode, leading to the value function failing to converge or even diverging, causing training to fail.

1.3 example

- Q-Learning without DQN.